Misc Notes on Single-Qubit Control

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- We set $\hbar = 1$ and write $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ for the Pauli vector, with $\sigma_\pm = (\sigma_x \pm i\sigma_y)/2$.
- For a mixed waveform $w(t) = a_x(t)\cos(\omega t) + a_y(t)\sin(\omega t)$, we use the complex-envelope convention $a(t) = a_x(t) + ia_y(t)$, such that $w(t) = [a(t)e^{-i\omega t} + a^*(t)e^{i\omega t}]/2$.
- We define the effective field $\mathbf{b}(t) = \omega_1 a_x(t)\hat{\mathbf{x}} + \omega_1 a_y(t)\hat{\mathbf{y}} + \Delta\omega\hat{\mathbf{z}}$, where $\Delta\omega$ is the resonance offset. The transceiver-frame Hamiltonian is $H_C(t) = -\mathbf{b}(t)\cdot\boldsymbol{\sigma}/2$.
- A rotation vector $\boldsymbol{\theta} = \theta\hat{\boldsymbol{\theta}}$ specifies the rotation $e^{-i\boldsymbol{\theta}\cdot\boldsymbol{\sigma}/2}$, with magnitude $\theta = |\boldsymbol{\theta}|$ equal to the rotation angle and direction $\hat{\boldsymbol{\theta}}$ along the rotation axis.
- Left and right matrix products: $\overrightarrow{\prod_{j=1}^n} B_j = B_1 \cdots B_n$ and $\overleftarrow{\prod_{j=1}^n} B_j = B_n \cdots B_1$.

1 Resonance-Offset Response of Shaped Pulses

- Consider a driven qubit with lab-frame Hamiltonian

$$\begin{aligned} H_L(t) &= -\frac{\omega_0}{2}\sigma_z - \omega_1\big(a_x(t)\cos(\omega t) + a_y(t)\sin(\omega t)\big)\sigma_x \\ &= -\frac{\omega_0}{2}\sigma_z - \frac{\omega_1}{2}\big(a(t)e^{-i\omega t} + a^*(t)e^{i\omega t}\big)\sigma_x, \end{aligned}$$

  where $a(t)$ is the dimensionless pulse envelope with $|a(t)| \le 1$.

- RWA in the transceiver frame gives

$$H_C(t) = -\frac{\Delta\omega}{2}\sigma_z - \frac{\omega_1}{2}\big(a(t)\sigma_- + a^*(t)\sigma_+\big) = -\frac{1}{2}\mathbf{b}(t)\cdot\boldsymbol{\sigma}.$$

- We look at the system's evolution at time $T \ll \omega_1^{-1}$. Go to the interaction frame of $-\Delta\omega\sigma_z/2$. This gives

$$\tilde{H}(t) = -\frac{\omega_1}{2}\big(a(t)e^{i\Delta\omega t}\sigma_- + a^*(t)e^{-i\Delta\omega t}\sigma_+\big).$$

  From the Dyson series,

$$\begin{aligned} \tilde{U}(T) &= \mathrm{Texp}\left[-i\int_0^T dt\,\tilde{H}(t)\right] = \mathbb{1} - i\int_0^T dt\,\tilde{H}(t) + \mathcal{O}\big(\|\tilde{H}\|_{\max}^2 T^2\big) \\ &= \mathbb{1} + i\frac{\omega_1}{2}\left(\int_0^T dt\,a(t)e^{i\Delta\omega t}\right)\sigma_- + i\frac{\omega_1}{2}\left(\int_0^T dt\,a(t)e^{i\Delta\omega t}\right)^*\sigma_+ + \mathcal{O}(\omega_1^2 T^2). \end{aligned}$$

- Now look at how the qubit's state deviates from $|0\rangle$ because of the pulse. Extending $a(t)$ to $\mathbb{R}$ by setting it to zero outside $[0, T]$, we have

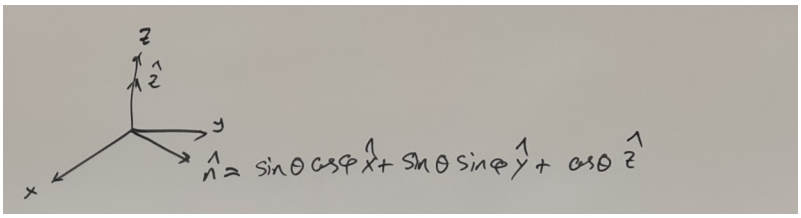
$$\boxed{\langle 1|\,\tilde{U}(T)\,|0\rangle \propto \int_{-\infty}^{\infty} dt\,a(t)e^{i\Delta\omega t} + \mathcal{O}(\omega_1^2 T^2).} \tag{1}$$

  Note that the transition probability $|\,\langle 1|\,\tilde{U}(T)\,|0\rangle\,|^2$ is the same in the transceiver frame, since the two frames only differ by a z-rotation.

- To first order in $\omega_1 T$, the qubit's response to the pulse is determined by its Fourier spectrum.

2 Rotations and State Transfer

- *Question:* What is the general single-qubit operation taking $\hat{\mathbf{z}}$ to an arbitrary state $\hat{\mathbf{n}} = \sin\theta\cos\phi\,\hat{\mathbf{x}} + \sin\theta\sin\phi\,\hat{\mathbf{y}} + \cos\theta\,\hat{\mathbf{z}}$.



A general $U \in \mathrm{SU}(2)$ has the form

$$U = \begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix},$$

where $\alpha, \beta \in \mathbb{C}$ with $|\alpha|^2 + |\beta|^2 = 1$. We want

$$U\,|0\rangle = e^{i\gamma}\left(\cos\frac{\theta}{2}\,|0\rangle + e^{i\phi}\sin\frac{\theta}{2}\,|1\rangle\right),$$

where $e^{i\gamma}$ is a global phase. This gives

$$\begin{pmatrix} \alpha \\ -\beta^* \end{pmatrix} = e^{i\gamma}\begin{pmatrix} \cos(\theta/2) \\ e^{i\phi}\sin(\theta/2) \end{pmatrix}, \qquad \alpha = e^{i\gamma}\cos\frac{\theta}{2}, \quad \beta = -e^{-i(\gamma+\phi)}\sin\frac{\theta}{2}.$$

Hence

$$\boxed{U = \begin{pmatrix} e^{i\gamma}\cos(\theta/2) & -e^{-i(\gamma+\phi)}\sin(\theta/2) \\ e^{i(\gamma+\phi)}\sin(\theta/2) & e^{-i\gamma}\cos(\theta/2) \end{pmatrix}, \qquad \gamma \in [0, 2\pi).} \tag{2}$$

- *Geometric interpretation:* find a rotation vector $\boldsymbol{\xi}$ such that $U = e^{-i\boldsymbol{\xi}\cdot\boldsymbol{\sigma}/2}$. Writing U in the Pauli basis,

$$\begin{aligned} U &= \cos\gamma\cos\frac{\theta}{2}\,\mathbb{1} + i\sin\gamma\cos\frac{\theta}{2}\,\sigma_z - i\cos(\gamma+\phi)\sin\frac{\theta}{2}\,\sigma_y + i\sin(\gamma+\phi)\sin\frac{\theta}{2}\,\sigma_x \\ &= \cos\gamma\cos\frac{\theta}{2}\,\mathbb{1} + i\left[\sin(\gamma+\phi)\sin\frac{\theta}{2}\,\hat{\mathbf{x}} - \cos(\gamma+\phi)\sin\frac{\theta}{2}\,\hat{\mathbf{y}} + \sin\gamma\cos\frac{\theta}{2}\,\hat{\mathbf{z}}\right]\cdot\boldsymbol{\sigma}. \end{aligned}$$

But

$$e^{-i\boldsymbol{\xi}\cdot\boldsymbol{\sigma}/2} = \cos\frac{\xi}{2}\,\mathbb{1} - i\sin\frac{\xi}{2}\,\hat{\boldsymbol{\xi}}\cdot\boldsymbol{\sigma}. \tag{3}$$

Comparing coefficients gives $\cos(\xi/2) = \cos\gamma\cos(\theta/2)$ and

$$\boxed{\begin{aligned} &\xi = 2\cos^{-1}\left(\cos\gamma\cos\frac{\theta}{2}\right), \\ &\hat{\boldsymbol{\xi}} = -\frac{\sin(\gamma+\phi)\sin(\theta/2)\,\hat{\mathbf{x}} - \cos(\gamma+\phi)\sin(\theta/2)\,\hat{\mathbf{y}} + \sin\gamma\cos(\theta/2)\,\hat{\mathbf{z}}}{\sqrt{1-\cos^2\gamma\cos^2(\theta/2)}}. \end{aligned}} \tag{4}$$

- Check that the axis is normalized:

$$\begin{aligned} |\hat{\boldsymbol{\xi}}|^2 &= \frac{\sin^2(\gamma+\phi)\sin^2(\theta/2) + \cos^2(\gamma+\phi)\sin^2(\theta/2) + \sin^2\gamma\cos^2(\theta/2)}{1-\cos^2\gamma\cos^2(\theta/2)} \\ &= \frac{\sin^2(\theta/2) + \sin^2\gamma\cos^2(\theta/2)}{1-\cos^2\gamma\cos^2(\theta/2)} = \frac{1 + (\sin^2\gamma - 1)\cos^2(\theta/2)}{1-\cos^2\gamma\cos^2(\theta/2)} = 1. \quad \checkmark \end{aligned}$$

3 The Pulse Flip Operation

- Consider a *resonant* control pulse on $[0, T]$. Here $\Delta\omega = 0$, so the effective field is $\mathbf{b}(t) = \omega_1 a_x(t)\hat{\mathbf{x}} + \omega_1 a_y(t)\hat{\mathbf{y}}$. The propagator is

$$U_C(T) = \mathrm{Texp}\left[\frac{i}{2}\int_0^T dt\, \mathbf{b}(t)\cdot\boldsymbol{\sigma}\right].$$

Define the pulse flip operation by

$$U_C^F = e^{-i\pi\sigma_z/2} U_C^\dagger e^{i\pi\sigma_z/2}. \tag{5}$$

Claim: This is the propagator of the flipped waveform $\mathbf{b}^F(t) = \mathbf{b}(T-t)$.

- *Proof:* Expand the original propagator in a Dyson series:

$$U_C(T) = \sum_{n=0}^{\infty}\left(\frac{i}{2}\right)^n \int_0^T dt_1 \cdots \int_0^{t_{n-1}} dt_n \prod_{j=1}^{n}\big(\mathbf{b}(t_j)\cdot\boldsymbol{\sigma}\big).$$

The $n = 0$ term is taken to be $\mathbb{1}$. Change variables to $t_j' = T - t_{n+1-j}$, so $t_n' = T - t_1$ and $t_1' = T - t_n$. This maps the original integration region $\mathcal{R} = \{(t_1, \ldots, t_n) \mid 0 \le t_n \le \cdots \le t_1 \le T\}$ to itself:

$$\mathcal{R}' = \{(t_1', \ldots, t_n') \mid 0 \le t_n' \le \cdots \le t_1' \le T\} = \mathcal{R}.$$

Therefore

$$\begin{aligned} U_C(T) &= \sum_{n=0}^{\infty}\left(\frac{i}{2}\right)^n \int_0^T dt_1' \cdots \int_0^{t_{n-1}'} dt_n' \prod_{j=1}^{n}\big(\mathbf{b}(T-t_j')\cdot\boldsymbol{\sigma}\big) \\ &= \left[\sum_{n=0}^{\infty}\left(-\frac{i}{2}\right)^n \int_0^T dt_1' \cdots \int_0^{t_{n-1}'} dt_n' \prod_{j=1}^{n}\big(\mathbf{b}^F(t_j')\cdot\boldsymbol{\sigma}\big)\right]^\dagger. \end{aligned}$$

Since $\mathbf{b}^F(t)\cdot\hat{\mathbf{z}} = 0$,

$$e^{i\pi\sigma_z/2}\big(\mathbf{b}^F(t)\cdot\boldsymbol{\sigma}\big)e^{-i\pi\sigma_z/2} = -\mathbf{b}^F(t)\cdot\boldsymbol{\sigma}.$$

Substituting this into each factor gives

$$\begin{aligned} U_C(T) &= \left[\sum_{n=0}^{\infty}\left(-\frac{i}{2}\right)^n \int_0^T dt_1' \cdots \int_0^{t_{n-1}'} dt_n' \prod_{j=1}^{n}\left(-e^{i\pi\sigma_z/2}\big(\mathbf{b}^F(t_j')\cdot\boldsymbol{\sigma}\big)e^{-i\pi\sigma_z/2}\right)\right]^\dagger \\ &= \left[e^{i\pi\sigma_z/2}\left(\sum_{n=0}^{\infty}\left(\frac{i}{2}\right)^n \int_0^T dt_1' \cdots \int_0^{t_{n-1}'} dt_n' \prod_{j=1}^{n}\big(\mathbf{b}^F(t_j')\cdot\boldsymbol{\sigma}\big)\right)e^{-i\pi\sigma_z/2}\right]^\dagger \\ &= e^{i\pi\sigma_z/2}\big(U_C^F(T)\big)^\dagger e^{-i\pi\sigma_z/2}. \end{aligned}$$

Rearranging gives $U_C^F(T) = e^{-i\pi\sigma_z/2}U_C^\dagger(T)e^{i\pi\sigma_z/2}$, as desired.

4 Virtual Z Gates

- Consider shifting the phase of a control pulse with complex envelope $a(t)$ by a constant φ to get $\tilde{a}(t) \equiv a(t)e^{-i\varphi}$, keeping the detuning $\Delta\omega$ fixed.

Claim: The propagator of the shifted pulse is related to the original propagator by

$$\boxed{\tilde{U}_C(t) = e^{i\varphi\sigma_z/2} U_C(t) e^{-i\varphi\sigma_z/2}.} \tag{6}$$

Proof: In the rotating frame, the pulse Hamiltonian is

$$H_C(t) = -\frac{1}{2}\mathbf{b}(t)\cdot\boldsymbol{\sigma} = -\frac{\Delta\omega}{2}\sigma_z - \frac{\omega_1}{2}\big[a(t)\sigma_- + a^*(t)\sigma_+\big].$$

Since $e^{i\varphi\sigma_z/2}\sigma_\pm e^{-i\varphi\sigma_z/2} = e^{\pm i\varphi}\sigma_\pm$, the shifted Hamiltonian is

$$\tilde{H}_C(t) = -\frac{\Delta\omega}{2}\sigma_z - \frac{\omega_1}{2}\big[a(t)e^{-i\varphi}\sigma_- + a^*(t)e^{i\varphi}\sigma_+\big] = e^{i\varphi\sigma_z/2} H_C(t) e^{-i\varphi\sigma_z/2}.$$

Hence,

$$\begin{aligned}\tilde{U}_C(t) &= \text{Texp}\left[-i\int_0^t dt'\,\tilde{H}_C(t')\right] = e^{i\varphi\sigma_z/2}\text{Texp}\left[-i\int_0^t dt'\,H_C(t')\right]e^{-i\varphi\sigma_z/2} \\ &= e^{i\varphi\sigma_z/2}U_C(t)e^{-i\varphi\sigma_z/2},\end{aligned}$$

as desired. Here, we used the fact that $\text{Texp}\left[\int_0^T dt\,SA(t)S^{-1}\right] = S\,\text{Texp}\left[\int_0^T dt\,A(t)\right]S^{-1}$ for any time-independent similarity transformation. To see this, consider the defining initial value problem for $V(t) \equiv \text{Texp}\left[\int_0^t dt'\,A(t')\right]$:

$$\dot{V}(t) = A(t)V(t),$$

with $V(0) = \mathbb{1}$. Define $W(t) = SV(t)S^{-1}$. Then

$$\dot{W}(t) = S\dot{V}(t)S^{-1} = SA(t)V(t)S^{-1} = \big(SA(t)S^{-1}\big)W(t),$$

with $W(0) = \mathbb{1}$. This is the defining initial value problem for $\text{Texp}\left[\int_0^t dt'\,SA(t')S^{-1}\right] = W(t)$.